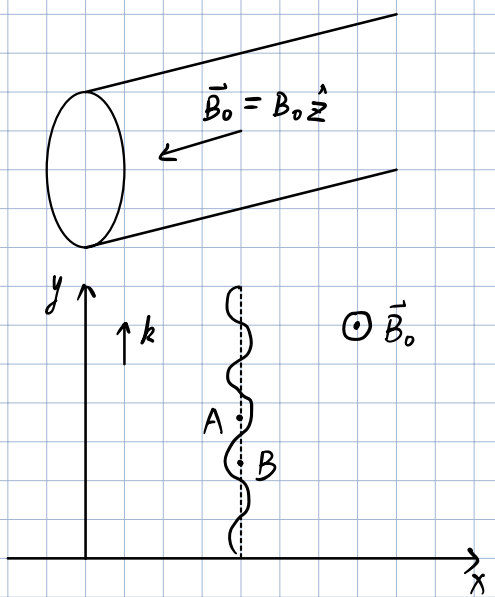
阻抗不稳定性.

$$\vec{k} = k_y \hat{y} + k_z \hat{z}$$



① $\vec{F} = kT\nabla n$,

$$\vec{v}_{i0} = \frac{\vec{F} \times \vec{B}_0}{e n_0 B_0^2} = \frac{kT_i \nabla n_0 \times \vec{B}_0}{e n_0 B_0^2}$$

$$\vec{v}_{e0} = \frac{-kT_e n_0'}{e n_0 B_0} \hat{y} = \frac{kT_i n_0'}{e n_0 B_0} \hat{y}$$

② 相速度.

$$\frac{\omega}{k} \simeq v_0 \qquad k_y \gg k_z.$$

$$\frac{\omega}{k_y}$$

$$\frac{n_1}{n_0} = \frac{e\phi_1}{KT_e} \qquad n_1 = n_0 \frac{e\phi_1}{KT_e}$$

电场与磁场耦合引起漂移.

$$\vec{v}_1 = \frac{\vec{E}_1 \times \vec{B}_0}{B_0^2} = \frac{-\nabla\phi_1 \times \vec{B}_0}{B_0^2} = i\vec{k}\phi_1 \times \vec{B}_0 / B_0^2$$

$$= \frac{ik_y \phi_1}{B_0} \hat{x}$$

③ 引导中心的连续性方程.

$$\frac{\partial n_1}{\partial t} + \vec{v}_1 \nabla n_0 = 0$$

假设 v_1 不随 x 变化.

$$\Rightarrow \qquad -i\omega n_1 + v_1 n_0' = 0$$

$$-i\omega n_1 = -v_1 n_0' = -i\omega \frac{n_0 e\phi_1}{k_B T_e}$$

$$\frac{\omega}{k} = -\frac{i\phi_1 n_0' k_B T_e}{B_0\, i n_0 e\phi_1} = \frac{iky\,\phi_1}{B_0} n_0' = -\frac{n_0'}{n_0}\frac{k_B T_e}{eB_0} \Rightarrow v_{e0}.$$

离子: $v_{1x} \neq \frac{E_1}{B_0}$. 修正 $\longrightarrow$ 密度与电势有相位差.

相互扰动 $\longrightarrow$ 越来越不稳定.

除非冷却至一定承度.